

Weekly Report #2-3

Benchmarking Stream Clustering Algorithms in River

Contributor	Contributor	Reporting period	08 Jun 2026 – 21 Jun 2026
Mentors	Advisors / Mentors	Phase	Phase 1 & 2 - Pipeline, Metrics & Algorithm Integration
Repository	github.com/user/repo	Report date	September 11, 2026

1. Executive Summary

Objective - Set up River and familiarize with River, implement core stream clustering evaluation metrics, implement foundational algorithms, and formulate an objective hyperparameter benchmarking methodology.

Status - DONE Ahead of Schedule

Progress Tracker

Table 1: Comparison between Proposal Schedule and Actual Progress (Weeks 2 & 3).

Phase / Week	Proposal Milestone	Actual Reality (Weeks 2-3)	Status	Completion
P1.2 (Week 2)	Study River algorithms & setup dev env	Installed River, explored framework & pipeline setup	DONE	100%
P2.1 (Week 3)	Establish core benchmark metrics	Implemented 9 metrics (ARI, NMI, AMI, Purity, RI, Silhouette, SSQ, DB, CH)	DONE	100%
P3.1 (Weeks 6-7)	Implement 1st algorithm group (early)	Implemented 10 algorithms (5 foundational + 5 exploratory)	DONE	100%
Strategic Plan	Roadmap alignment	Prioritized foundational algorithms with sound theory/experiments	DONE	100%
Methodology	Benchmark design	Researched LODO framework [1] & proposed Relative Parameter LODO [2]	DONE	100%

Takeaway. Core pipeline setup and evaluation metrics are complete. Algorithm implementation and benchmark methodology formulation are substantially ahead of the original proposal schedule.

Overall: Weeks 2-3/18 - ahead of schedule.

2. Completed Tasks

C1. River Pipeline Setup and Evaluation Metrics Integration (Phases 1 & 2)

- Installed River and established a unified pipeline execution framework to streamline streaming model execution, evaluation, and algorithm integration.
- Implemented and wrapped comprehensive clustering evaluation metrics directly compatible with River streams:
 - **External metrics:** Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), Adjusted Mutual Information (AMI), Purity, Rand Index (RI).
 - **Internal metrics:** Silhouette Coefficient, Sum of Squared Errors (SSQ/SSE), Davies–Bouldin (DB) Index, Calinski–Harabasz (CH) Index.

C2. Stream Clustering Algorithm Implementations and Prioritization Strategy (Phase 3)

- Implemented 10 streaming clustering algorithms:
 - **Foundational, standard, and widely cited:** DStream, HPStream, Birch, RepStream, SVstream.
 - **Exploratory/other models:** DDStream, MrStream, DCUstream, improved_Birch, SOSstream.
- **Implementation Strategy:** Prioritize foundational algorithms first because they possess rigorous theoretical foundations and solid, reproducible experimental setups in original papers, making them ideal for verification. Conversely, other algorithms often present ambiguous algorithmic formulations and questionable experimental reproducibility.

C3. Research on Hyperparameter Benchmarking for Stream Clustering (Methodology)

- **Current State & Challenges:**
 - *Overfitting:* Most literature reports hyperparameters manually tuned specifically on the authors' chosen datasets. On different benchmarks, performance drops drastically or exhibits unexpected runtime/RAM spikes.
 - *Unfair Comparison:* Cerqueira et al. [1] emphasize that heuristic parameter choices or tuning on test data creates severe evaluation bias.
 - *Core Objective:* Identify hyperparameter configurations that perform robustly and objectively across diverse datasets, rather than finding an isolated "best" parameter for a single dataset.
- **Reference Baseline - Leave-One-Dataset-Out (LODO):**
 - *Mechanism (Cerqueira et al. [1]):* Use $n - 1$ datasets to search for optimal configurations (via random search) and evaluate on the remaining held-out dataset. Keeps tuning independent from test data.
 - *Advantages:* Model-agnostic, evaluates wide parameter spaces, straightforward, reproducible.
 - *Drawbacks:* High computational cost, enormous search space, reliance on chance rather than domain insight.
 - *Fatal Limitation for Clustering:* Standard LODO fails because static absolute parameters (e.g., fixed ϵ) lack consistent geometric meaning across datasets with varying dimensionality, scales, and cluster counts (e.g., KDD99 with 23 clusters vs. Shuttle with 2 clusters). No single static absolute value works universally.
- **Proposed Solution - Purposeful LODO with Relative Parameters:**
 - (a) **Tune Key Parameters Only:** Focus exclusively on 1–2 pivotal parameters (or more depending on algorithm) cluster structure and granularity, for examples:
 - *Centroid (KMeans, etc.):* k (number of clusters).

- *Density* (*DenStream*, etc.): ϵ , `clustering_threshold` (micro-cluster radius and merge/split criterion).
 - *Grid* (*DStream*, etc.): `grid_width` (cell density threshold).
 - *Hierarchical* (*BIRCH*, etc.): `threshold` (CF-tree branching threshold).
 - *Graph* (*RepStream*, etc.): α (representative point density).
- (b) **Use Relative Parameters via Data Statistics** (`param = f(stats)`): Tune relative scaling coefficients instead of static values:
- $\epsilon = c \cdot d$, where d is median pairwise distance or average k -NN distance.
 - `grid_width` = $\text{range}(X)/g$, where g is target divisions per dimension.
 - `threshold` = $c \cdot \sigma$, where σ is mean standard deviation across dimensions.
 - k and α tied to class counts or sample mode estimates.
- (c) **Core Benefit:** Supported by the precedent of Fan et al. [2], a single scaling coefficient c automatically adapts to data density (larger ϵ for sparse data, smaller for dense data), eliminating the flaw of enforcing rigid absolute values across heterogeneous datasets.

3. Reproducing the SVStream Experiments

We reproduced the experimental study of SVStream [3] (Figures 3–18) with our River implementation and benchmark framework. Every difference in the input shows up in the results, so this section first states which inputs can follow the paper and which cannot, then how the open ones were fixed, and only then compares the results figure by figure.

3.1 Inputs: What Can and Cannot Follow the Paper

- **Ring-Ball and Smileface-Twomoons cannot match the paper exactly** (Figure 3 and every figure that uses them). The data were never released and the paper’s figures have no axes, so guessing coordinates by eye would not make the input closer to the paper, only more arbitrary. We keep what the paper states (sizes, classes, noise ratio, order of the phases) and calibrate the coordinate scale *once* against the best q of the paper’s Figure 15 (Section 3.2). Matching them exactly would require the data from the authors.
- **The normalisation of the real streams is not stated in the paper.** It is chosen by the same one-off calibration and recorded explicitly.
- **KDD CUP’99 and Forest CoverType are sampled.** This run uses the first 10,000 records of each stream in every figure (Figures 14–16 and 18), whereas the paper streams all 494,020 and 581,012 records. The benchmark supports the full streams, but they would take tens of hours. Figure 18 additionally keeps its earlier setting (running z-score, $q = 0.02 / 0.15$, unscaled synthetic streams). Real-stream numbers are therefore not comparable with the paper’s.

3.2 Calibration of the Open Inputs

Coordinate scale. A Gaussian kernel of width q on data scaled by s behaves like qs^2 on the unscaled data, so a single q sweep on the unscaled stream fixes $s = \sqrt{q^*/q_{\text{paper}}}$. Ring-Ball peaks at $q^* = 8$ against the paper’s 9 ($s = 0.943$); Smileface-Twomoons peaks at $q^* = 50$ against 11 ($s = 2.13$). Because the scale is chosen by maximising the average Rand index at the paper’s defaults, the synthetic panels of Figure 15 and the average Rand index of Figures 5–6 and 11–12 are not independent evidence.

Normalisation. At the paper’s own q , we compute $q \cdot \max \|\mathbf{x}_i - \mathbf{x}_j\|^2$ on the first chunk for each candidate normalisation, since Section 4.4.2 of the paper asks for q slightly larger than $1/\max \|\mathbf{x}_i - \mathbf{x}_j\|^2$, and discard normalisations under which almost every kernel value is 0 or 1.

Table 2: Inputs of the reproduction compared with the paper.

Input	Paper	This run	Same?
Parameters and grids	Table 1 (M, q, C); $\delta = 0.6, \eta = 1, \zeta = 2$; sweep grids of Figs. 13–17	Used as given in Figs. 3–17	Yes
Synthetic streams	Not released; figures without axes	Stated properties kept; coordinate scale calibrated once	No
KDD CUP’99	494,020 records, 34 attributes, “one outlier removed”	First 10,000 records in original order (the removed outlier, index 346,366, lies outside them)	No
Forest CoverType	581,012 records, 10 attributes	First 10,000 records in original order	No
Normalisation	Not stated	Calibrated once: KDD online z-score, CoverType min-max	No
Figure 13 grid	$C \in [0.26, 0.34]$	13a: the paper’s grid; 13b: $C \in [0.010, 0.022]$, where the box binds	13a only
Figure 18	Full streams, paper’s q	10,000 records, online z-score, $q = 0.02$ / 0.15, unscaled synthetic streams	No
Baselines, platform	StrAP included; unstated implementation on a Core2 Duo PC; peak memory usage	River baselines without StrAP; Python; model size	No

On KDD CUP’99 only the online z-score puts this product near one (0.99); on CoverType only min-max leaves a usable kernel (27.6; with the others more than 99% of the kernel values are ≈ 0). The choice uses kernel statistics only, not the Rand index.

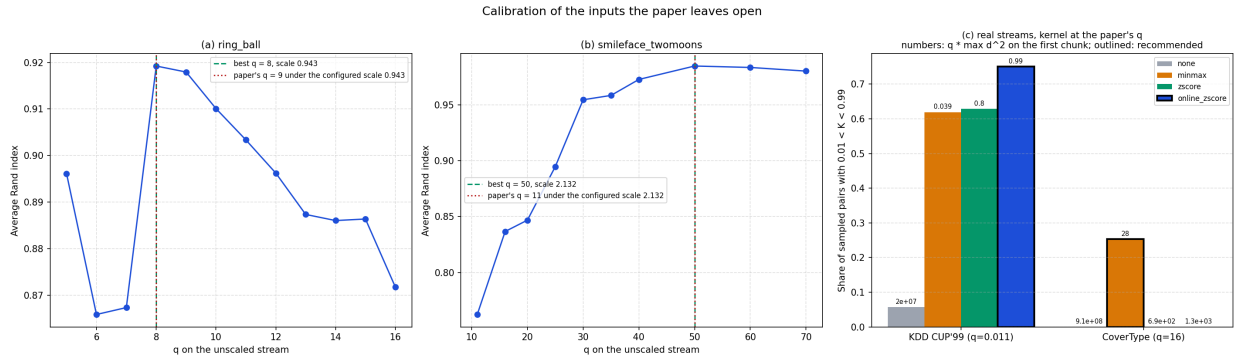
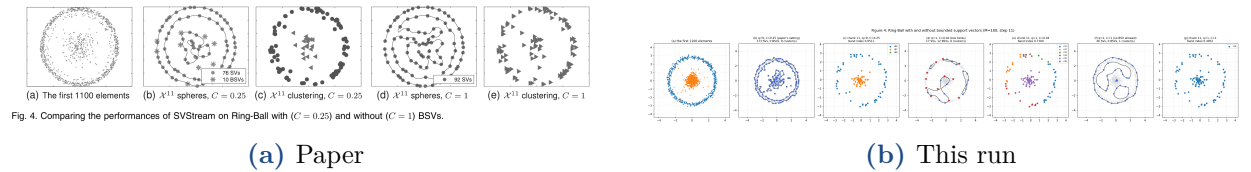


Figure 1: Calibration. (a, b) Rand index of q sweeps on the unscaled synthetic streams: dashed, the best q ; dotted, the paper’s q under the chosen scale. (c) Share of sampled pairs with an informative kernel value at the paper’s q for each normalisation; numbers give $q \cdot \max \|\mathbf{x}_i - \mathbf{x}_j\|^2$ on the first chunk, outlined bars the chosen normalisation.

3.3 Results and Analysis

All figures below were produced with the inputs of Table 2. Each figure is followed by the analysis of how it differs from the paper.

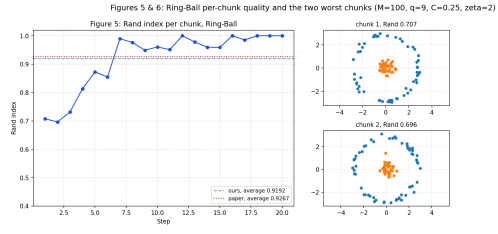
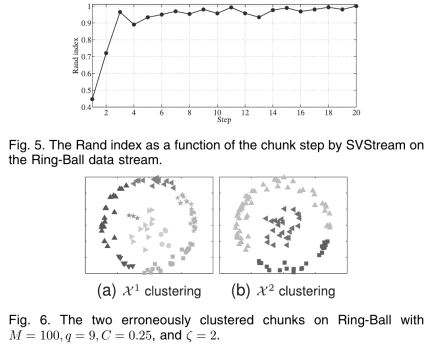


Analysis. Only the failure at $C = 1$ reproduces: the two classes merge into one cluster as in the paper. At the paper’s $C = 0.25$ we get no BSV at all (173 SVs, 8 clusters), although the chunk is still clustered well (RI 0.951). The reason is the box constraint of Eq. (5): with $\sum_i \beta_i = 1$ and

Table 3: Paper findings vs. this run. [†]Value on which the coordinate scale was calibrated.

Fig.	Paper	This run	Match
3	Two synthetic streams	Re-created, scaled by 0.943 and 2.13	Input
4	$C = 0.25$: 76 SVs, 10 BSVs, classes separated; $C = 1$: one cluster	$C = 0.25$: 173 SVs, 0 BSVs , 8 clusters (chunk RI 0.951); $C = 1$: one cluster (0.495)	Partly
5–6	Avg. RI 0.9267; low RI only in chunks 1–2	Avg. RI 0.9192 [†] ; RI below 0.9 in chunks 1–6	Yes
7–10	53–126 SVs+BSVs, 3–14 BSVs	138→506 SVs, 0 BSVs; chunk RI 0.96–1.00, the converging eyes merged at step 70	Partly
11–12	Avg. RI 0.9857; 65/70 chunks > 0.95	Avg. RI 0.9844 [†] ; 63/70 chunks > 0.95	Yes
13	Detection 82–99%, false alarms < 6%; both rise as ζ and C fall and as q rises	13a: 0% everywhere. 13b: ζ and C trends match, q trend reversed; detection mostly 8–34%	No
14	RI stable in M except on Ring-Ball; time grows with M ; < 200 SVs+BSVs; 12–230 KB	RI stable; time <i>falls</i> with M ; 110–874 SVs+BSVs; 18–181 KB	Partly
15	Best $q = 9$ (Ring-Ball), 11 (Smileface); KDD ≈ 0.66 , CoverType ≈ 0.71	Best $q = 11$ (0.923) and 12 (0.986) [†] ; KDD 0.97–1.00; CoverType flat at 0.33	Partly
16	Clear optimum in C on the synthetic streams; CoverType best at $C = 0.2$	Flat, or noisy without a trend on Ring-Ball	No
17	Stable for $\delta \in [0.5, 0.6]$ and $\eta \in [0.9, 1.0]$, lower for small δ and η ; (0.6, 1.0) chosen	Default at or within 0.002 of the best (Ring-Ball 0.919; Smileface 0.984 vs. 0.986); η has no effect, no drop for small δ and η	Partly
18	SVStream best on all streams, smallest memory	Best on Ring-Ball and KDD; behind baselines on unscaled Smileface and on CoverType	Partly

$\beta_i \leq C$, a sphere holds at most $\lfloor 1/C \rfloor$ BSVs, i.e. 4 at $C = 0.25$, yet the paper shows 10 BSVs here and 14 at $C = 0.3$ (Figure 8). Its C is therefore probably scaled differently. Implemented literally, the box only binds once C is lowered, e.g. to 0.04 (12 BSVs).



(a) Paper (b) This run
Figure 3: Figures 5–6, Rand index per chunk on Ring-Ball and the worst chunks.

Analysis. The average Rand index is close to the paper’s (0.919 vs. 0.927), but it is the value the scale was calibrated on. The early chunks are poor in both, which the paper attributes to the sparse first chunks leaving gaps within the clusters; ours stay below 0.9 for six chunks instead of two. The two chunk panels are coloured with the model at the end of the stream, not at their own step, which is why they look correct although these chunks scored 0.71 and 0.70.

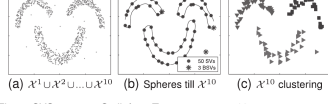


Fig. 7. SVStream on Smileface-Twomoons: step 10.

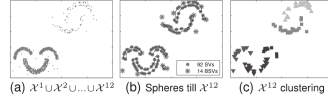


Fig. 8. SVStream on Smileface-Twomoons: step 12.

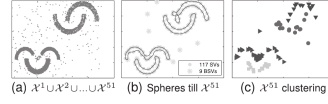


Fig. 9. SVStream on Smileface-Twomoons: step 51.

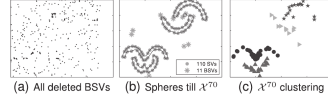


Fig. 10. SVStream on Smileface-Twomoons: step 70.

(a) Paper

(b) This run

Figure 4: Figures 7–10, Smileface-Twomoons at steps 10, 12, 51 and 70: stream so far, spheres, and clustering of the chunk.

Analysis. The chunk Rand index stays high at every step (0.96–1.00), but the clusterings are not as clean as the paper’s: at step 12 the newly emerged moons are still split into several small clusters (0.956), and at step 70 the two converging eyes form one cluster (0.971). The difference is the summary. Since no BSV is created at the paper’s C (see Figure 4), BSV decay never removes noise: noise points stay as isolated SVs and the summary grows from 138 to 506 SVs, whereas the paper keeps 53 to 126 SVs+BSVs and deletes the noise as BSVs (Figure 10a).

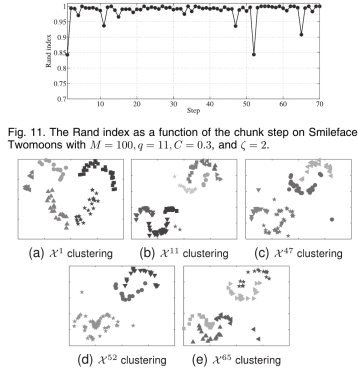

 Fig. 11. The Rand index as a function of the chunk step on Smileface-Twomoons with $M = 100$, $q = 11$, $C = 0.3$, and $\zeta = 2$.

 Fig. 12. All the five mistakenly clustered chunks on Smileface-Twomoons with $M = 100$, $q = 11$, $C = 0.3$, $\zeta = 2$.

(a) Paper

(b) This run

Figure 5: Figures 11–12, Rand index per chunk on Smileface-Twomoons and the five worst chunks.

Analysis. This figure now matches the paper: average RI 0.984 vs. 0.986 and 63 vs. 65 chunks above 0.95. The weakest chunks are chunk 1 (0.87), chunk 7 (0.92) and chunks 65–68 (down to 0.91), where the eyes merge. The average is the calibration target and not independent evidence, but the Rand index stays at 0.98 over every M (Figure 14) and every C (Figure 16), which the calibration did not tune, while the same algorithm on the unscaled stream scores only 0.762 (Figure 18). The earlier gap on this stream therefore came mostly from the data scale, i.e. a kernel too wide for our coordinates.

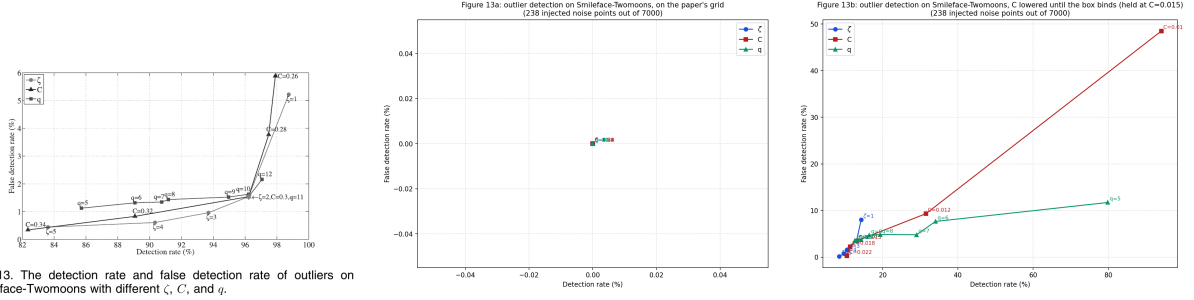


Fig. 13. The detection rate and false detection rate of outliers on Smileface-Twomoons with different ζ , C , and q .

(a) Paper

(b) 13a: paper's grid

(c) 13b: C where the box binds

Figure 6: Figure 13, outlier detection rate vs. false detection rate on Smileface-Twomoons for varying ζ , C and q .

Analysis. On the paper's grid (13a) no BSV exists, so no point is ever declared an outlier and every rate is 0%. With C lowered to where the box binds (13b), a lower ζ and a lower C raise both rates, as in the paper, but a larger q lowers them, and the detection rate stays at 8–34% except at the extremes $C = 0.010$ and $q = 5$ (paper: 82–99% with under 6% false alarms). We have not established why the mechanism behaves differently once it does engage.

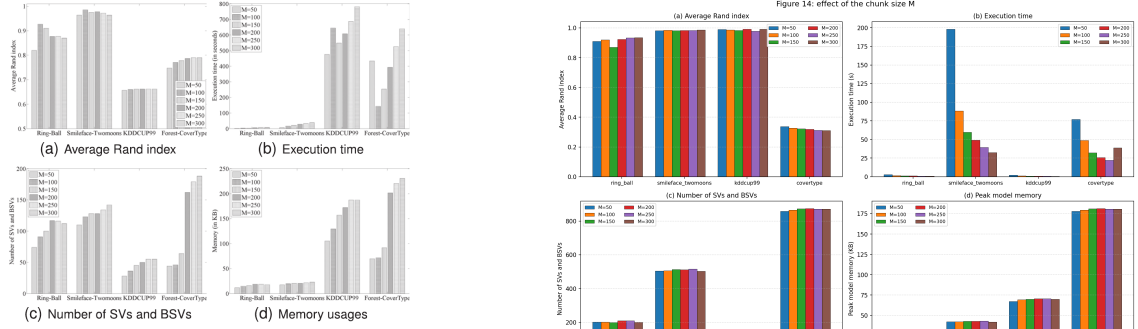


Fig. 14. Average Rand index, execution time, number of SVs and BSVs, and memory usages versus chunk size M .

(a) Paper

(b) This run

Figure 7: Figure 14, average Rand index, execution time, number of SVs and BSVs, and memory versus chunk size M .

Analysis. As in the paper, the Rand index is stable in M , and memory is of the same order (18–181 KB vs. 12–230 KB) and higher on KDD CUP'99 than on the 2-D streams; we measure the model size, not the paper's peak memory usage. Two panels differ. Runtime falls as M grows, because a smaller M means more QP solves over the same stream, whereas the paper's runtime grows with M . The summary exceeds the paper's bound of 200 (503–516 SVs on Smileface-Twomoons, about 870 on CoverType), because without BSVs nothing removes the noise (Figures 7–10).

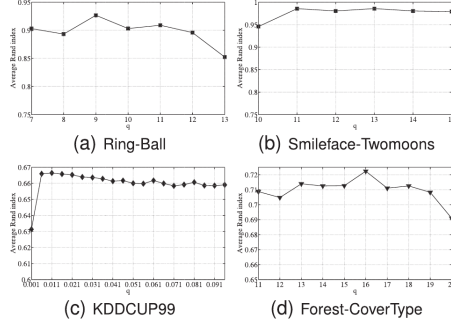


Fig. 15. Average Rand index versus Gaussian kernel parameter q on the four testing streams.

(a) Paper

(b) This run

Figure 8: Figure 15, average Rand index versus the kernel width q on the paper's grids.

Analysis. The synthetic panels stay close to the paper, partly by calibration: Ring-Ball ranges over 0.85–0.92 (paper 0.85–0.93) but peaks at $q = 11$ rather than 9, and Smileface-Twomoons is flat at 0.983–0.986 with its best at $q = 12$ (paper 11). The real panels are not comparable. On KDD CUP'99 the index sits at 0.97–1.00 and falls slowly with q (paper ≈ 0.66), which points to 10,000-record prefix chunks dominated by very few classes rather than to good clustering. On CoverType it is flat at 0.33 with a cluster detection rate of zero, consistent with a single cluster per chunk. A preliminary check outside the figure pipeline, on the first 30,000 records with min-max scaling computed over the whole stream, gave 0.52, so both the stream length and the records the scaling is computed on may matter; only a full-stream run can settle it.

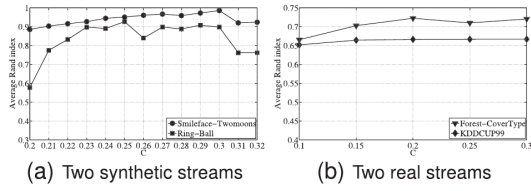


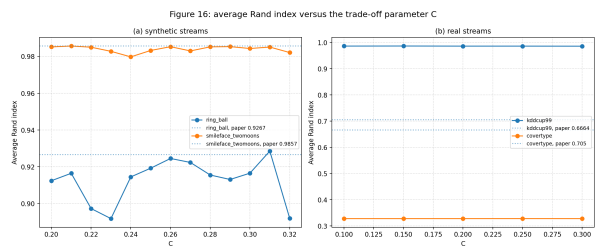
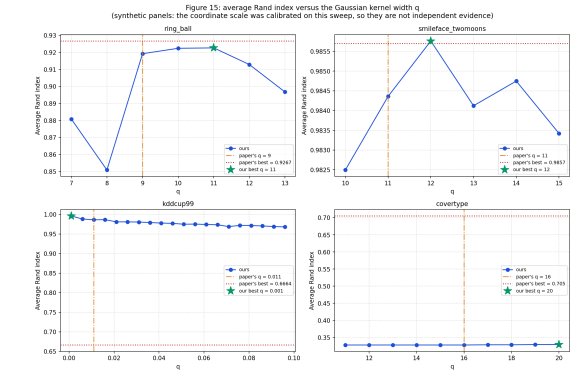
Fig. 16. Average Rand index versus tradeoff parameter C on the four testing streams.

(a) Paper

(b) This run

Figure 9: Figure 16, average Rand index versus the trade-off parameter C on the paper's grids.

Analysis. None of the paper's optima appears (Ring-Ball at $C = 0.25$, Smileface-Twomoons at 0.3, CoverType at 0.2). Across $C \in [0.1, 0.32]$ the box of Eq. (5) practically never binds (Figure 4), so C has no systematic effect: Smileface-Twomoons stays at 0.980–0.986, KDD CUP'99 at 0.986–0.987 and CoverType at 0.328, while Ring-Ball fluctuates between 0.89 and 0.93 without a trend.



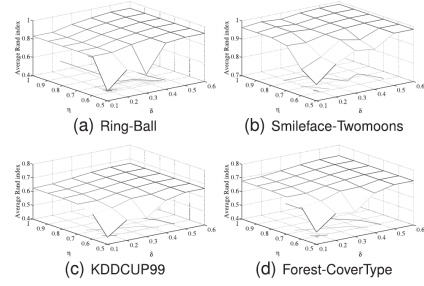
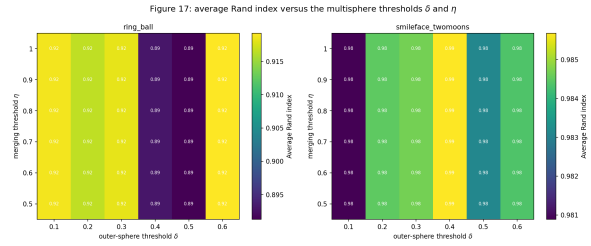


Fig. 17. Average Rand index versus multisphere parameters, i.e., δ and η .



(a) Paper

(b) This run

Figure 10: Figure 17, average Rand index versus the multisphere thresholds δ and η (this run: the two synthetic streams only).

Analysis. As in the paper, the default $(\delta, \eta) = (0.6, 1.0)$ is at or next to the best cell: it is the best on Ring-Ball (0.919) and within 0.002 of the best on Smileface-Twomoons (0.984 vs. 0.986 at $\delta = 0.4$). The paper’s sensitivity does not reproduce, however. Its surfaces drop to about 0.4–0.7 when both δ and η are small, whereas here η has no effect at all (every row of each heat map is identical) and δ moves the Rand index by at most 0.03 (Ring-Ball falls to 0.89 at $\delta = 0.4$ –0.5). We have not established why the merging threshold is inert. The benchmark only runs the two synthetic streams, so the paper’s KDD CUP’99 and CoverType panels are not reproduced.

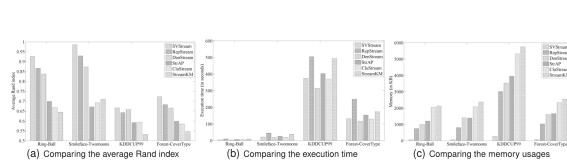
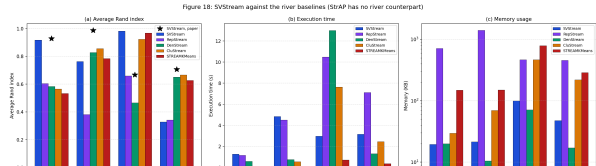


Fig. 18. Comparing the performances of SVStream with five algorithms.

(a) Paper

(b) This run

Figure 11: Figure 18, Rand index, execution time and memory against River baselines, with the unchanged Figure 18 setting (unscaled synthetic streams, 10,000-record real prefixes).



Analysis. SVStream wins clearly on Ring-Ball (0.918 vs. at most 0.60) and leads on KDD CUP’99 (0.982), but trails on Smileface-Twomoons (0.762, behind CluStream, DenStream and STREAMKMeans) and is last on CoverType (0.328). The Smileface-Twomoons result is the unscaled stream this figure keeps; at the calibrated scale SVStream reaches 0.984 (Figures 11–12). SVStream has the smallest memory only on Ring-Ball; DenStream is smaller on the other three streams. The comparison also differs from the paper’s through its baselines (River implementations, no StrAP, DenStream’s ϵ rescaled to the data), the Python implementation, and the memory metric.

Takeaway. Only the input parameters can follow the paper exactly; the synthetic streams can only be re-created and scaled, and the real streams here are 10,000-record samples. With the scale calibrated, Smileface-Twomoons and Ring-Ball come close to the paper’s clustering quality, but the BSV mechanism of Eq. (5) never engages at the paper’s C , so outlier detection, the C study and the summary size still differ from the paper.

4. Issues & Blockers

[No major blockers encountered during Weeks 2–3]

5. Next Week’s Plan

[Reserved for Next Week's milestones]

6. References & Links

Code produced in Weeks 2-3

Ref	Description	Link
Module 1	Pipeline setup and Evaluation Metrics (ARI, NMI, Silhouette, etc.)	metrics/river_eval
Module 2	Foundational & Experimental Clustering Algorithms	algorithms/stream_models
Module 3	Hyperparameter Benchmarking Harness (Relative LODO)	benchmarks/relative_lodo

Literature and Documentation

References

- [1] V. Cerqueira, H. M. Gomes, M. Heyden, B. Pfahringer, and A. Bifet, "A Framework for Evaluating and Benchmarking Concept Drift Detection Methods," in *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '26)*, 2026. arXiv:2606.07789.
- [2] X. Fan, Y. Yue, P. Sarkar, and Y. X. R. Wang, "On Hyperparameter Tuning in General Clustering Problems," in *Proceedings of the 37th International Conference on Machine Learning (ICML 2020)*, PMLR 119, pp. 2965–2975, 2020. URL: [proceedings.mlr.press/v119/fan20b.html](#).
- [3] C.-D. Wang, J.-H. Lai, D. Huang, and W.-S. Zheng, "SVStream: A Support Vector-Based Algorithm for Clustering Data Streams," *IEEE Transactions on Knowledge and Data Engineering*, vol. 25, no. 6, pp. 1410–1424, 2013. DOI: [10.1109/TKDE.2011.263](#).